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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:            0.0
Usage of /:             5.1% of 48.91GB
Memory usage:          5%
Swap usage:            0%
Processes:             139
Users logged in:       1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: Aquisap, John Paul L.
Student ID: 2024300588
Course/Year/Section: BSIT 3-B
Date Submitted: 02/08/2026
Instructor: Sir. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Linux
Virtualization Software (VirtualBox/VMware)	VirtualBox
Ubuntu Server Version	ubuntu-26.04-live-server-amd64
VM Name	Lab1_Aquisap
RAM Allocated	4
Storage Allocated	8000
Network Mode	(Internal Network,'labnetwork')
SSH Client Used	Secure Shell Client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	<u>8000</u> MB

Hard Disk	<u>100</u> GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

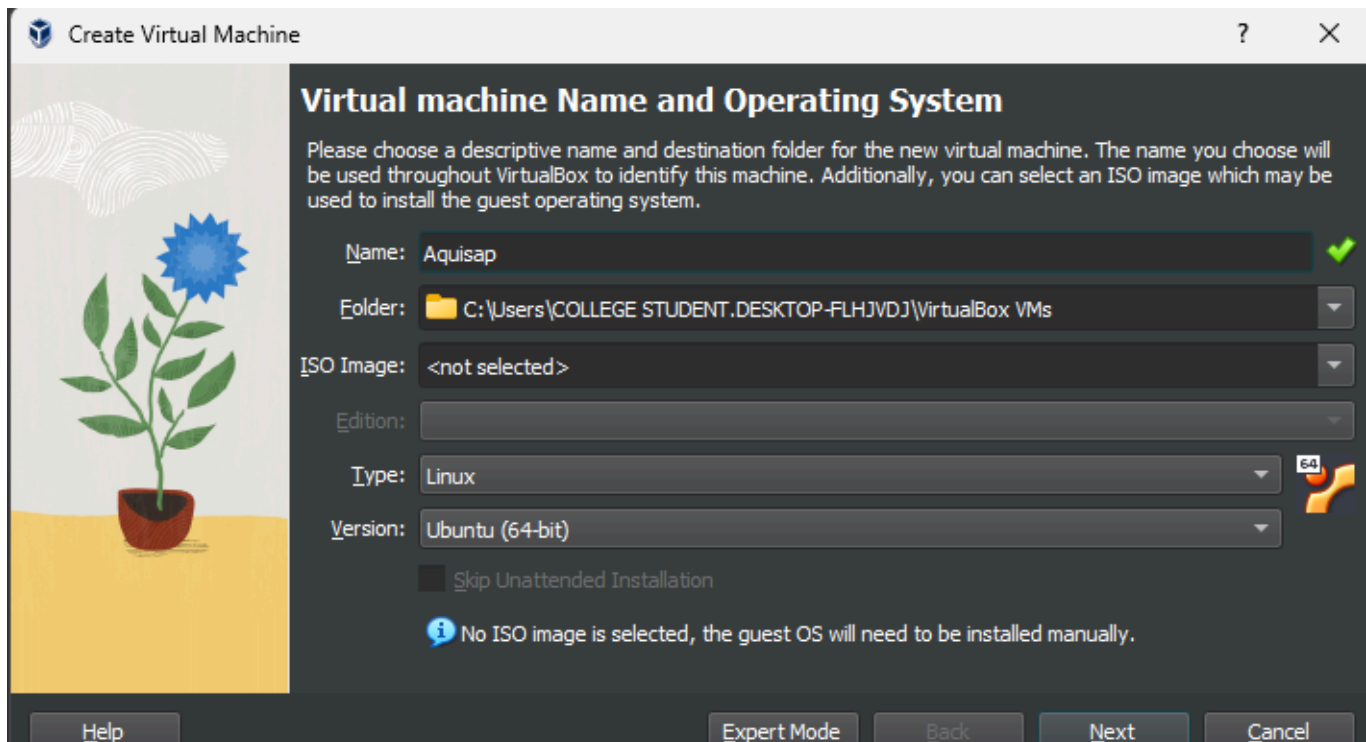
Server IP Address: 192.168.50.10

Network Interface: Internal Network

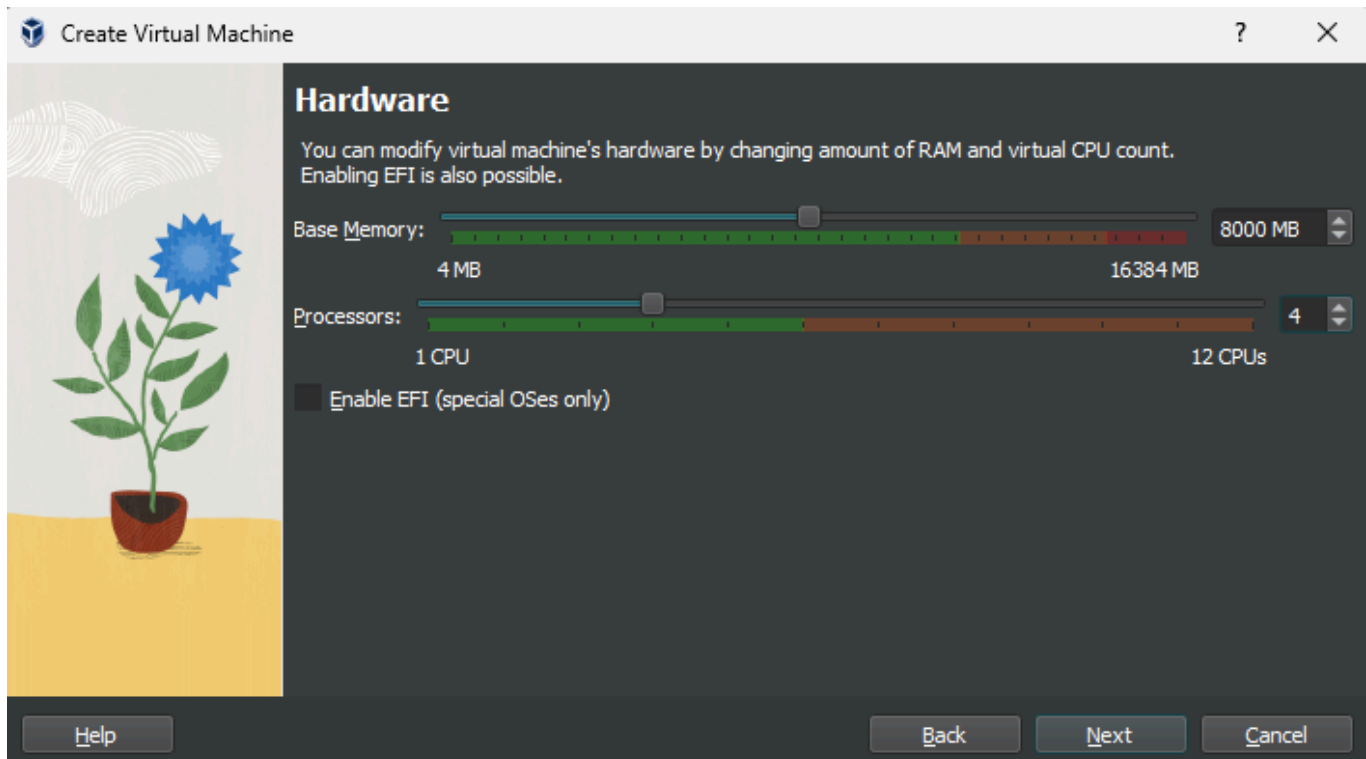
6. REMOTE CONNECTION

Connection Details:

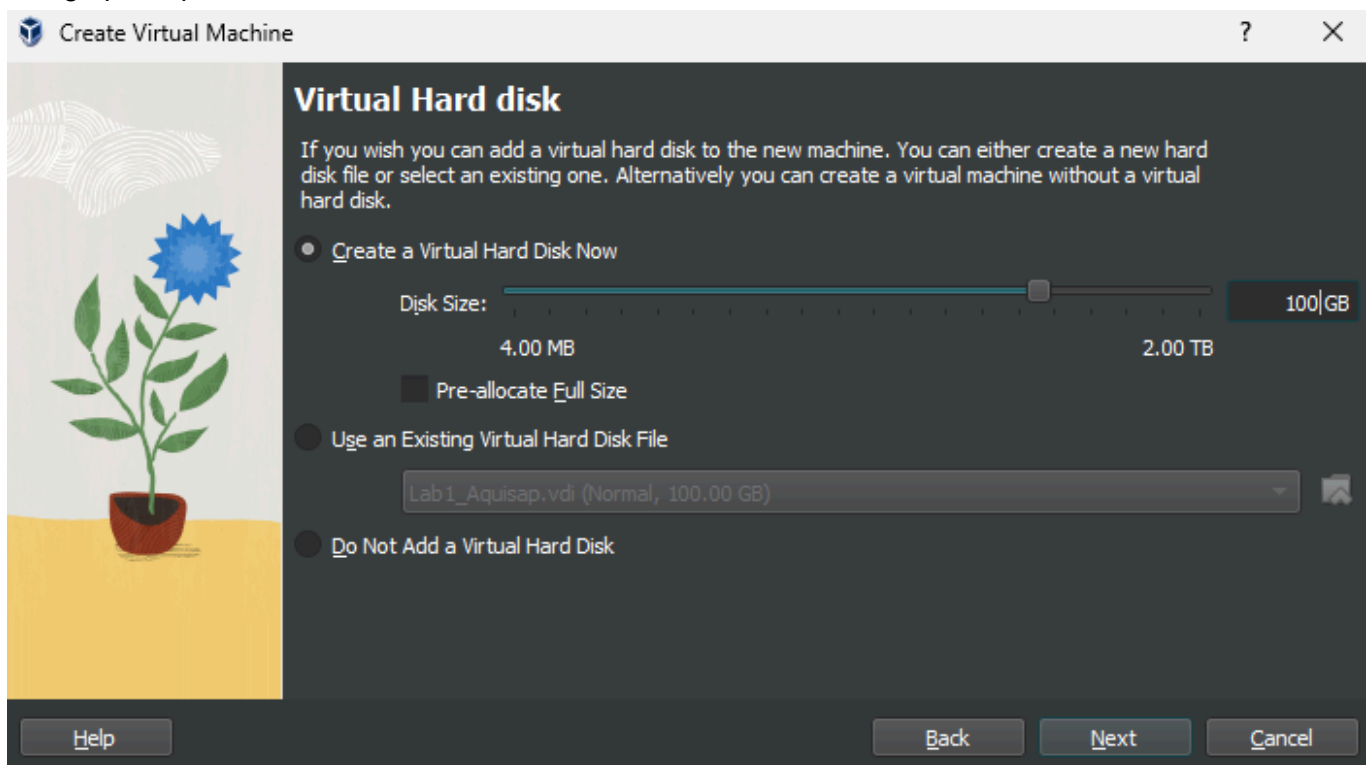
Parameter	Value
Server IP	192.168.50.10
Username	Aquisap
SSH Port	Internal Network



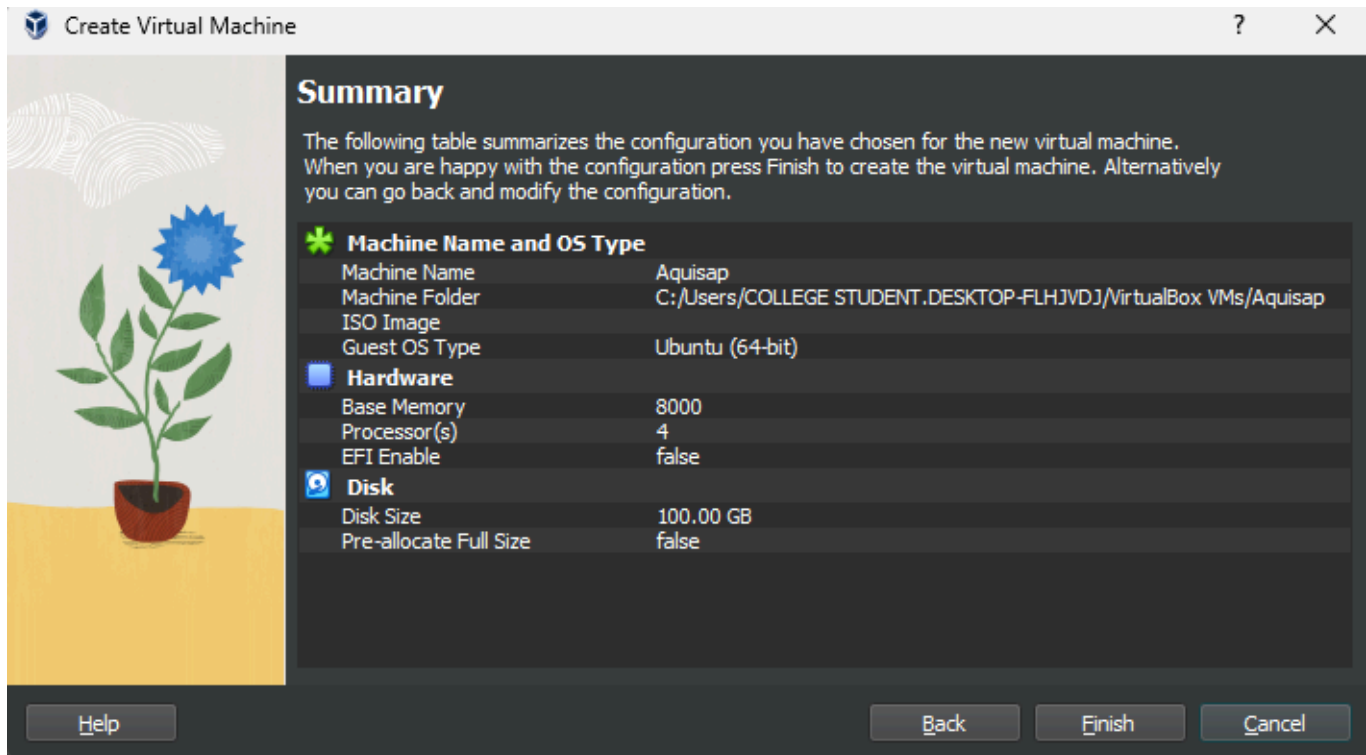
Start of installing Ubuntu Server



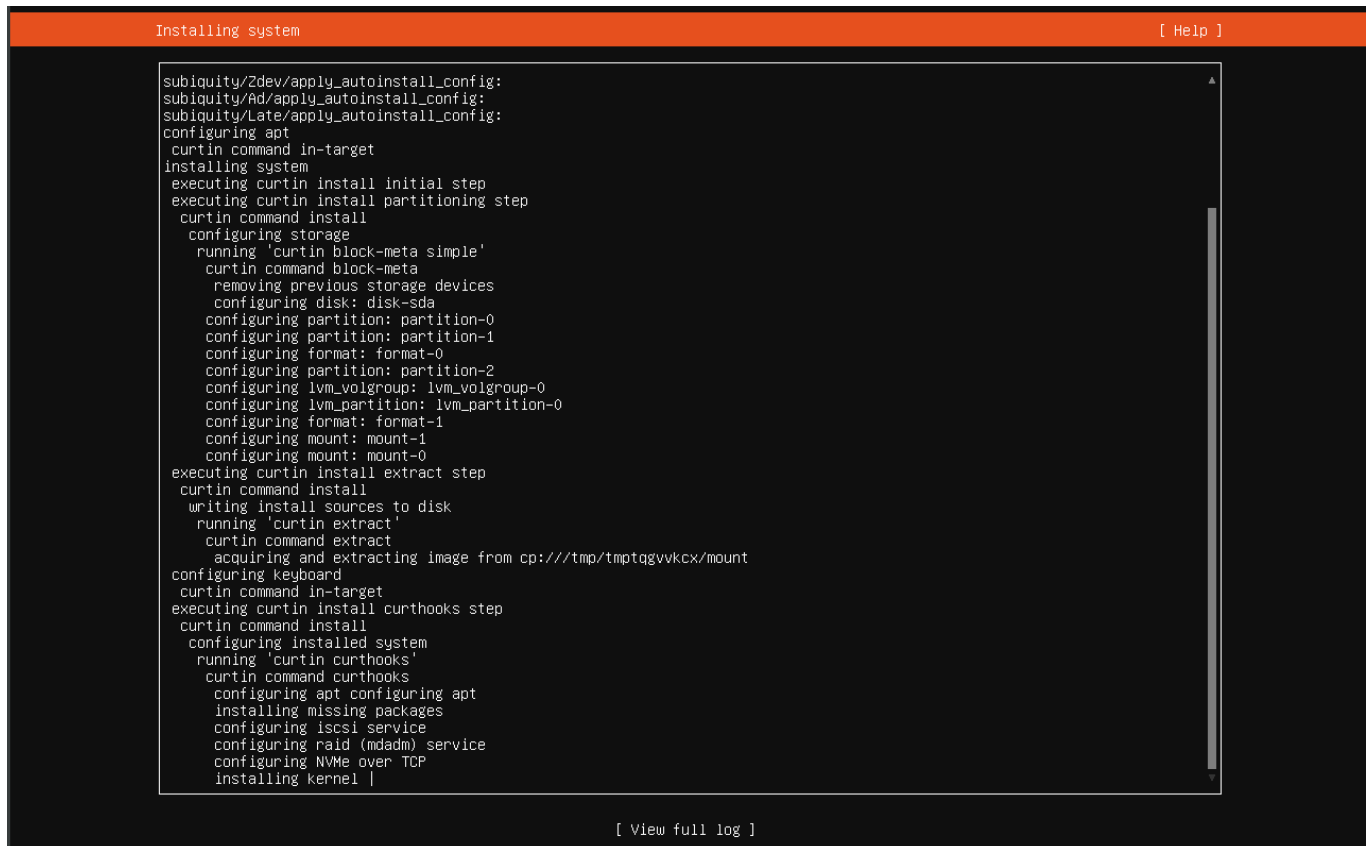
Filling up all specs needed



Filling up all specs needed



Final page in setup



Start installing ubuntu server

```
Aquisap [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

[ 14.551231] cloud-init[1489]: |+.B .
[ 14.556421] cloud-init[1489]: |.0 0 . .
[ 14.556498] cloud-init[1489]: | + * 0 + 0
[ 14.556521] cloud-init[1489]: | + * E 0 0
[ 14.556542] cloud-init[1489]: | B B = S
[ 14.556563] cloud-init[1489]: | = + = 0
[ 14.556587] cloud-init[1489]: |*+= 0
[ 14.556605] cloud-init[1489]: |=== .
[ 14.556626] cloud-init[1489]: |+= ..
[ 14.556652] cloud-init[1489]: +----[SHA256]-----+
[ 14.556675] cloud-init[1489]: Generating public/private ecdsa key pair.
[ 14.556700] cloud-init[1489]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 14.556732] cloud-init[1489]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 14.556766] cloud-init[1489]: The key fingerprint is:
[ 14.556787] cloud-init[1489]: SHA256:5HHdvAJpoVeV035HhSWERRmTSL8mIdugyP6mIhHdt+M root@aquisap
[ 14.556818] cloud-init[1489]: The key's randomart image is:
[ 14.556840] cloud-init[1489]: +---[ECDSA 256]-----+
[ 14.556872] cloud-init[1489]: | . .BXB+|
[ 14.556904] cloud-init[1489]: | . =.*0.|
[ 14.556949] cloud-init[1489]: | . . + * ..++|
[ 14.556974] cloud-init[1489]: | . . + * . .0+|
[ 14.556999] cloud-init[1489]: | . S .0....0|
[ 14.563223] cloud-init[1489]: | . . .0. =.. .|
[ 14.563703] cloud-init[1489]: | . 0.... 0 0|
[ 14.563958] cloud-init[1489]: | . . . E 0|
[ 14.564403] cloud-init[1489]: | . .0+.|
[ 14.564697] cloud-init[1489]: +----[SHA256]-----+
[ 14.564868] cloud-init[1489]: Generating public/private ed25519 key pair.
[ 14.565121] cloud-init[1489]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 14.565252] cloud-init[1489]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 14.565475] cloud-init[1489]: The key fingerprint is:
[ 14.565629] cloud-init[1489]: SHA256:tYx8oq1mDhLLdvECY5E3YtF5YhyFG2CqD03KDxDogmM root@aquisap
[ 14.565804] cloud-init[1489]: The key's randomart image is:
[ 14.566044] cloud-init[1489]: +--[ED25519 256]--+
[ 14.566240] cloud-init[1489]: | ++.=|
[ 14.566489] cloud-init[1489]: |.o o0 .|
[ 14.566789] cloud-init[1489]: |0.*.0= .|
[ 14.567061] cloud-init[1489]: |==+. . + .|
[ 14.567440] cloud-init[1489]: |0E* . S +|
[ 14.567648] cloud-init[1489]: |0*+= 0 0 0|
[ 14.567800] cloud-init[1489]: | *.+ 0 .|
[ 14.567917] cloud-init[1489]: | . 0 oo.|
[ 14.568027] cloud-init[1489]: | +0|
[ 14.568133] cloud-init[1489]: +----[SHA256]-----+

Ubuntu 26.04 LTS aquisap tty1

aquisap login: _
```

follow the process

```
Aquisap [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

System load: 0.1          Processes:           154
Usage of /: 15.0% of 47.9GB Users logged in:      0
Memory usage: 3%          IPv4 address for enp0s3: 10.0.2.15
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jpaquisap@aquisap:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
  Active: active (running) since Wed 2026-09-02 06:34:21 UTC; 2min 34s ago
  Invocation: dff7d936e24743b78cc079d8397661a4
  TriggeredBy: • ssh.socket
  Docs: man:sshd(8)
        man:sshd_config(5)
  Process: 5564 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 5566 (sshd)
  Tasks: 1 (limit: 5974)
  Memory: 1.6M (peak: 2.7M)
  CPU: 22ms
  CGroup: /system.slice/ssh.service
          └─5566 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 06:34:21 aquisap systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:34:21 aquisap sshd[5566]: Server listening on 0.0.0.0 port 22.
Sep 02 06:34:21 aquisap sshd[5566]: Server listening on :: port 22.
Sep 02 06:34:21 aquisap systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
jpaquisap@aquisap:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
jpaquisap@aquisap:~$ sudo apt install_
```

install and activate ssh

```
Aquisap [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

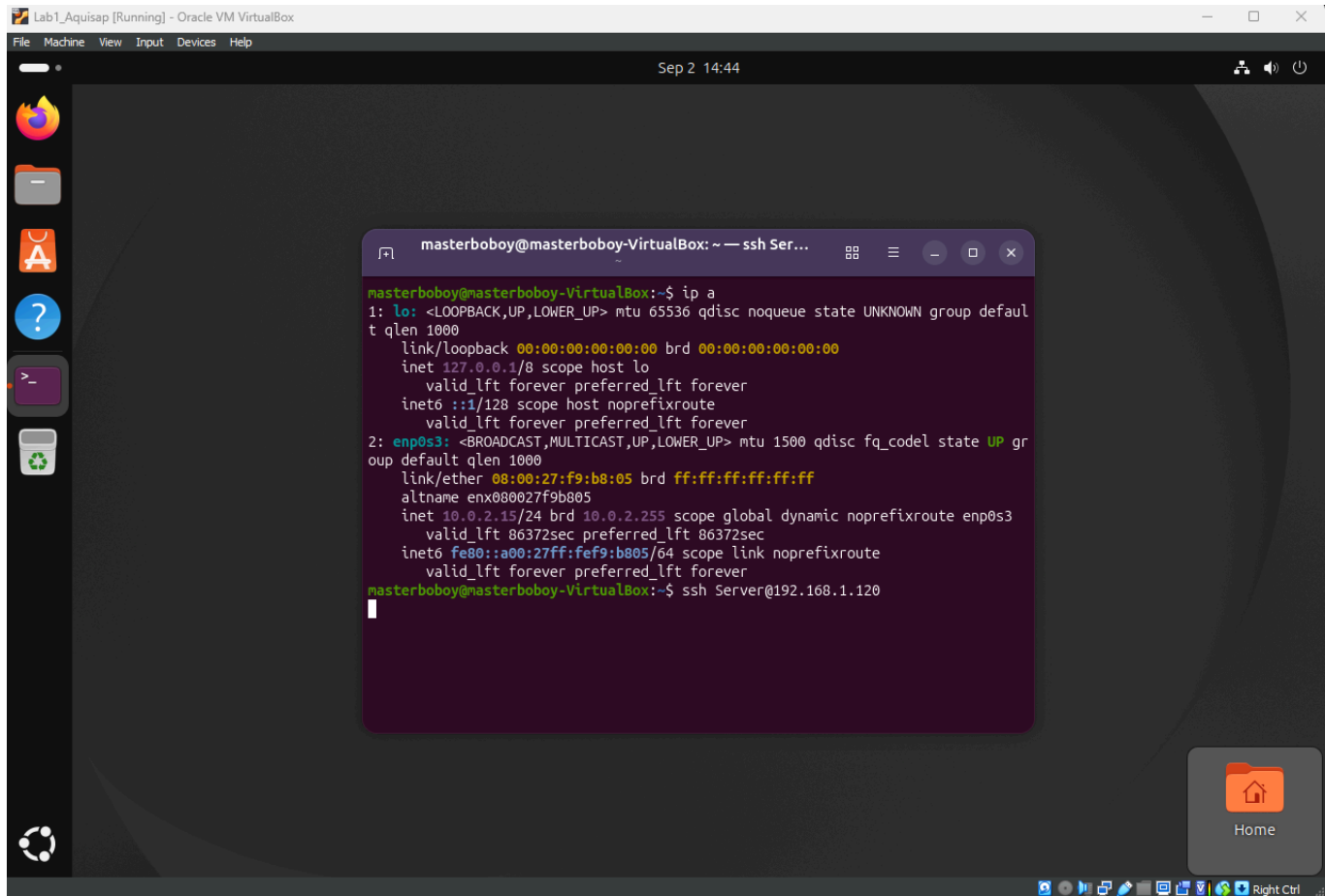
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jpaquisap@aquisap:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 06:34:21 UTC; 2min 34s ago
   Invocation: dff7d936e24743b78cc079d0397661a4
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
          man:sshd_config(5)
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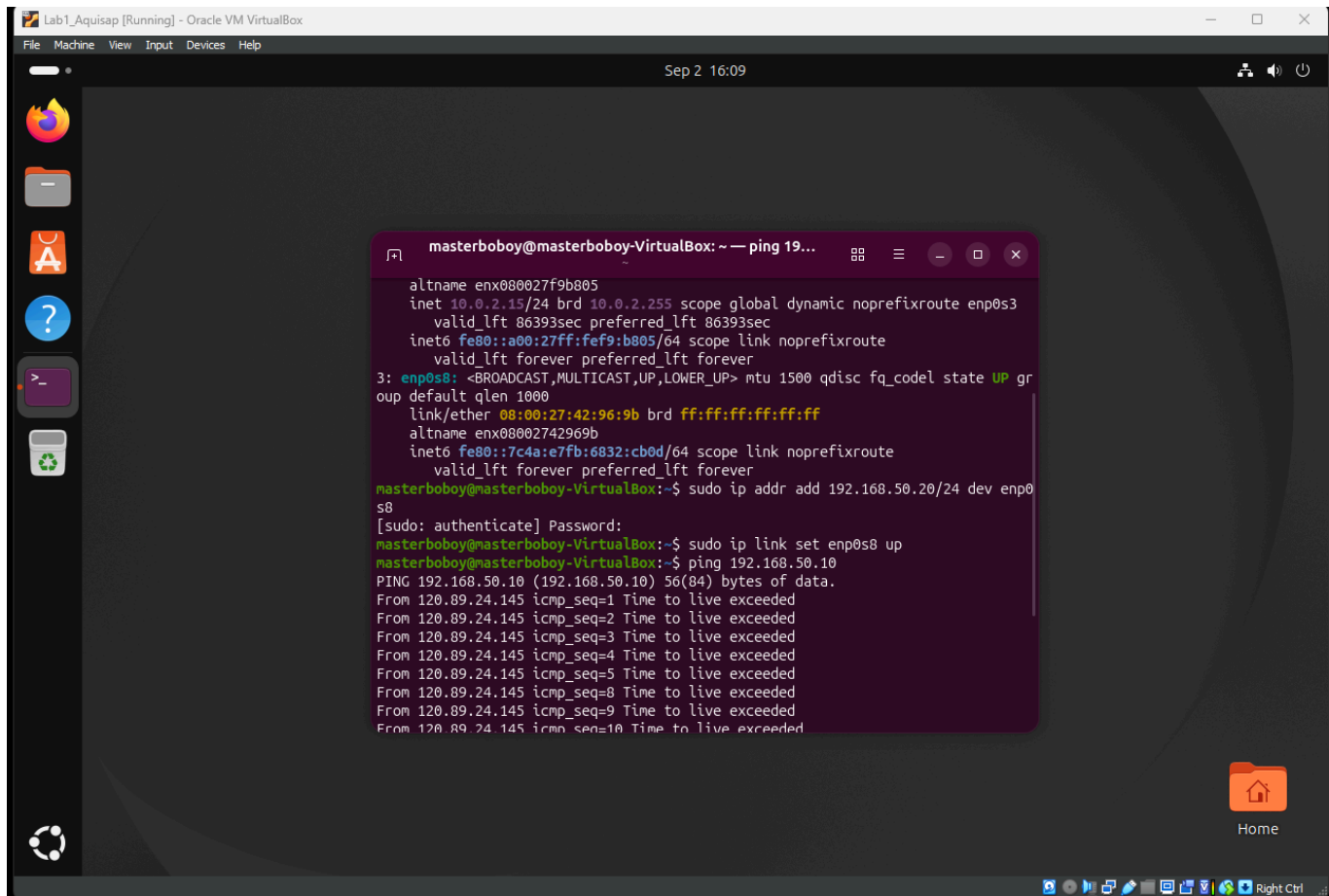
Sep 02 06:34:21 aquisap systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:34:21 aquisap sshd[5566]: Server listening on 0.0.0.0 port 22.
Sep 02 06:34:21 aquisap sshd[5566]: Server listening on :: port 22.
Sep 02 06:34:21 aquisap systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
jpaquisap@aquisap:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
jpaquisap@aquisap:~$ sudo apt install openssh-server-y
Error: Unable to locate package openssh-server-y
jpaquisap@aquisap:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55
jpaquisap@aquisap:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
jpaquisap@aquisap:~$ sudo systemctl start ssh
sudo: 'systemctl': command not found
jpaquisap@aquisap:~$ sudo systemctl start ssh
jpaquisap@aquisap:~$
```

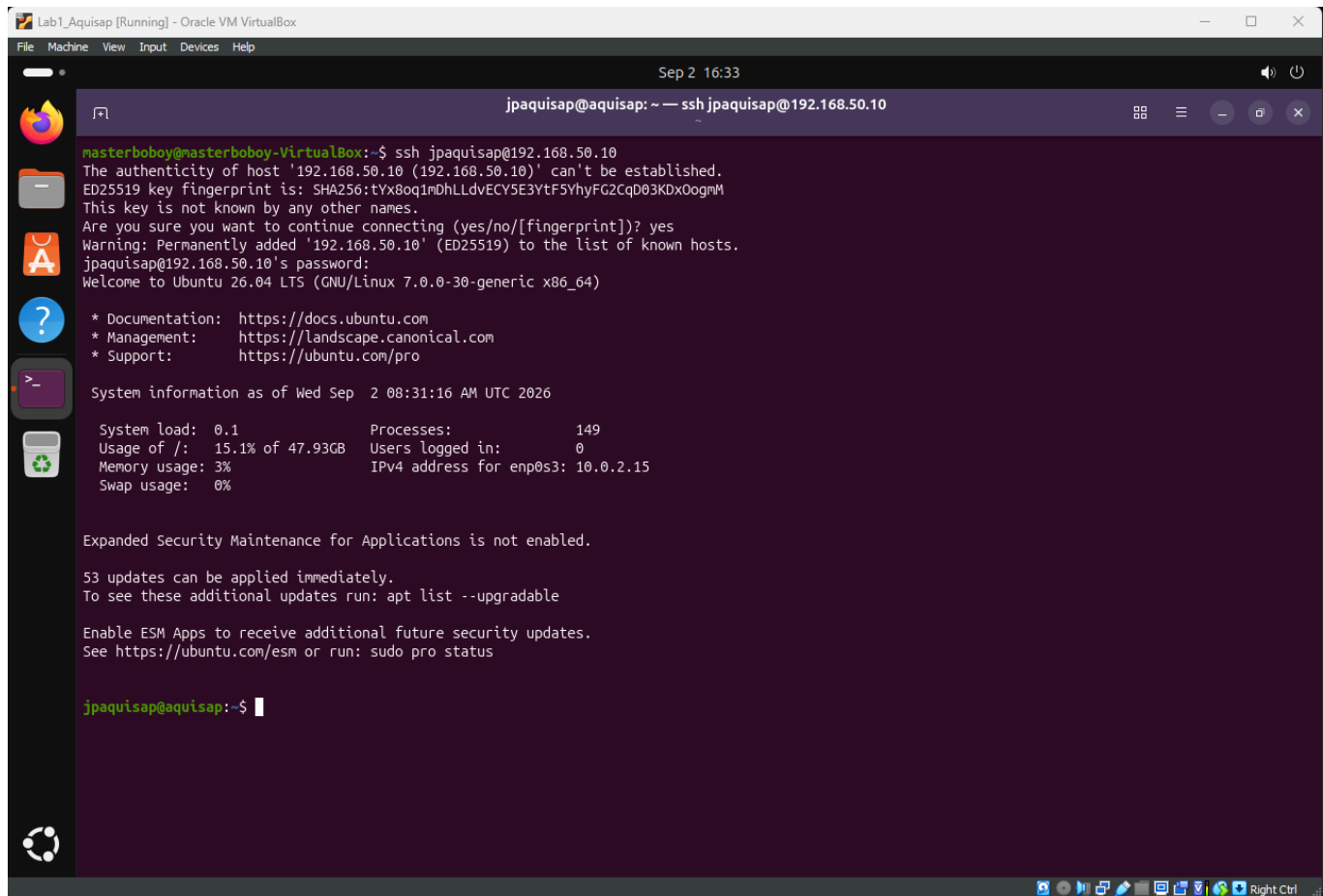
try to connect it to your ubuntu desktop



input your ip address into your desktop terminal to connect



encountered an error where my ip address could not connect to desktop



```
Lab1_Aquisap [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Sep 2 16:33
jpaquisap@aqisap: ~ — ssh jpaquisap@192.168.50.10

masterboboy@masterboboy-VirtualBox:~$ ssh jpaquisap@192.168.50.10
The authenticity of host '192.168.50.10 (192.168.50.10)' can't be established.
ED25519 key fingerprint is: SHA256:tYx8oq1mDhLLdvECY5E3YtF5YhyFG2CqD03KDX0ogmM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.50.10' (ED25519) to the list of known hosts.
jpaquisap@192.168.50.10's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 08:31:16 AM UTC 2026

System load:  0.1               Processes:            149
Usage of /:   15.1% of 47.93GB   Users logged in:     0
Memory usage: 3%               IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

jpaquisap@aqisap:~$
```

Finally solved it and finished the activity